



Gen AI and Agentic AI in Production

Week 2
Day 5

4 Week Journey to Prod-Grade

Week 4: Agentic AI in Production

Multi-Agent
Architecture

Capstone:
Agents on
AWS

Capstone:
Frontend

Observability
Monitoring
Security

Agentic AI
Platforms

Week 3: Azure, GCP, Data Engineering, Vectors, MCP

Cyber AI on
Azure with
MCP

Cyber AI on
GCP with
MCP

Amazon
SageMaker

Data Ingest
S3 Vectors

Data Pipes
with MCP

Week 2: AI Platform Engineering on AWS

AWS
Architecture

The Digital
Twin Mk 2

Amazon
Bedrock

Terraform

GitHub
Actions

Week 1: SaaS Generative AI in Production

First Gen AI
Deployment

Frontend &
Backend

Auth &
Subscription

Healthcare
SaaS App

Deploy to
AWS

Recap: Terraform

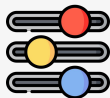
Provider



A vendor like AWS

A plugin to make Terraform apply to that provider

Variable



Parameters

Configurable settings that affect your deployment

Resource



The building block

Each AWS service to be created is described with a resource

State



Record of resources

Maps the configuration to actual resources

Output



Results of deployment

Like, actual CloudFront distribution URLs

Workspace



Separated state

Isolated state for the same configuration

GitHub Actions for CICD

Automate deployments using Github tooling

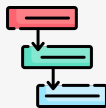
Actions



Platform

Run scripts in response to actions like doing a git push

Workflows



Orchestration

YAML files stored in .github/workflows

Jobs



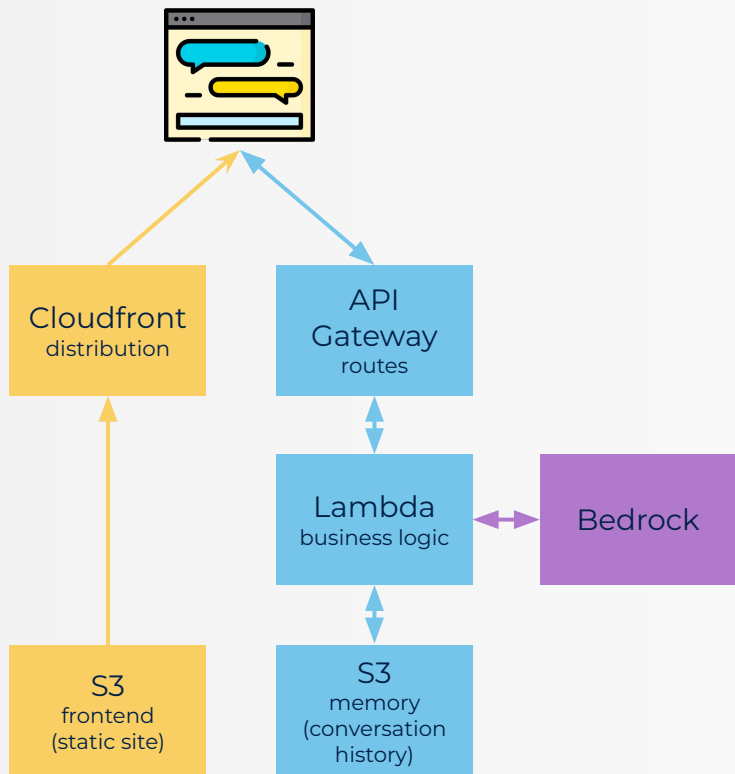
Execution

Collection of steps that run on a 'Runner' VM that

GitHub Actions can be useful beyond CICD

And there are other platforms for CICD, like Jenkins

Deployment Architecture



Recap

1. We built a digital twin app with memory
2. We deployed it to **AWS** using the AWS console
3. We built a **Serverless architecture**, using **Lambda**, **S3**, **API gateway**, **Cloudfront** distribution
4. We wrote **Terraform** scripts to create and destroy 3 environments
5. Then we automated deployment with **GitHub actions**

oh and yes..

..we **used Amazon Bedrock to connect with Nova...** 😂

ASSIGNMENT: Add functionality to the app using tools, like we did on the Agentic AI course. Perhaps use an S3 bucket for Q&A and to record contact details!

HALF WAY MOMENT!!



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50%
Complete